Lattice Design Workshop:
OPA code for electron beam optics in accelerators

Gabriele Benedetti
gbenedetti@cells.es

Download OPA

In a Windows PC or laptop, download the latest OPA version from

https://opa-code.github.io/

Have a look at the documentation.

Specific instructions to develop two types of cell (FODO and DBA) will be given in class.

OPA code web page

https://opa-code.github.io

Search

OPA Lattice Design Code

Please visit the **OPA repository** github.com/opa-code for source code, releases and documentation.

Direct links

Download the latest **Windows executable**: github.com/opa-code/opa-4/releases/latest

PDF documention

- OPA code documentation (v.4.2.4)
- "inside OPA" (v.4.2.4): the physics part.
- OPA user guide (v.4.047)
- Tutorial (v.4.033)

Note change in version numbering: 4.063 was renamed 4.1.0

Additional documentation

- Example lattice file for the tutorial.
- Files (zip) for using current export (unit opacurrents).

Contact: Andreas Streun

OPA input/output files

- *filename.opa*: lattice text file. To define the lattice elements and sequences of elements to build a lattice
- *filename.wmf*: plot export as Windows metafile (figures to be inserted in documents…)
- *filename.txt* and *filename.out*: plot data export as text file

But OPA is mainly designed for **visual and interactive use**.

Emittance and focusing as a function of bending angle and lattice style

In this workshop you will asked to observe how emittance and optics functions (betas, dispersion…) change as a function of the bending angle θ of rectangular dipoles.

Here you have some useful formulae seen in class for the equilibrium emittance and the focusing properties of rectangular dipoles:

$$\varepsilon_0 \approx C_q \gamma^2 \theta^3 F$$

F is a factor depending on the type of cell (FODO, DBA…) and its horizontal phase advance (90°, 137°…):

Lattice style	F
90° FODO	$2\sqrt{2}$
137° FODO	1.2
Double-bend achromat (DBA)	$\frac{1}{4\sqrt{15}}$
Multi-bend achromat	$\frac{1}{12\sqrt{15}}\left(\frac{M+1}{M-1}\right)$
TME	$\frac{1}{12\sqrt{15}}$

The horizontal and vertical transfer matrices for a rectangular bending (as in FODO to be build with OPA) are:

$$M_{H\ rectangular} = \begin{pmatrix} 1 & \rho sin\theta & \rho(1-\cos\theta) \\ 0 & 1 & \sin\theta \\ 0 & 0 & 1 \end{pmatrix}$$

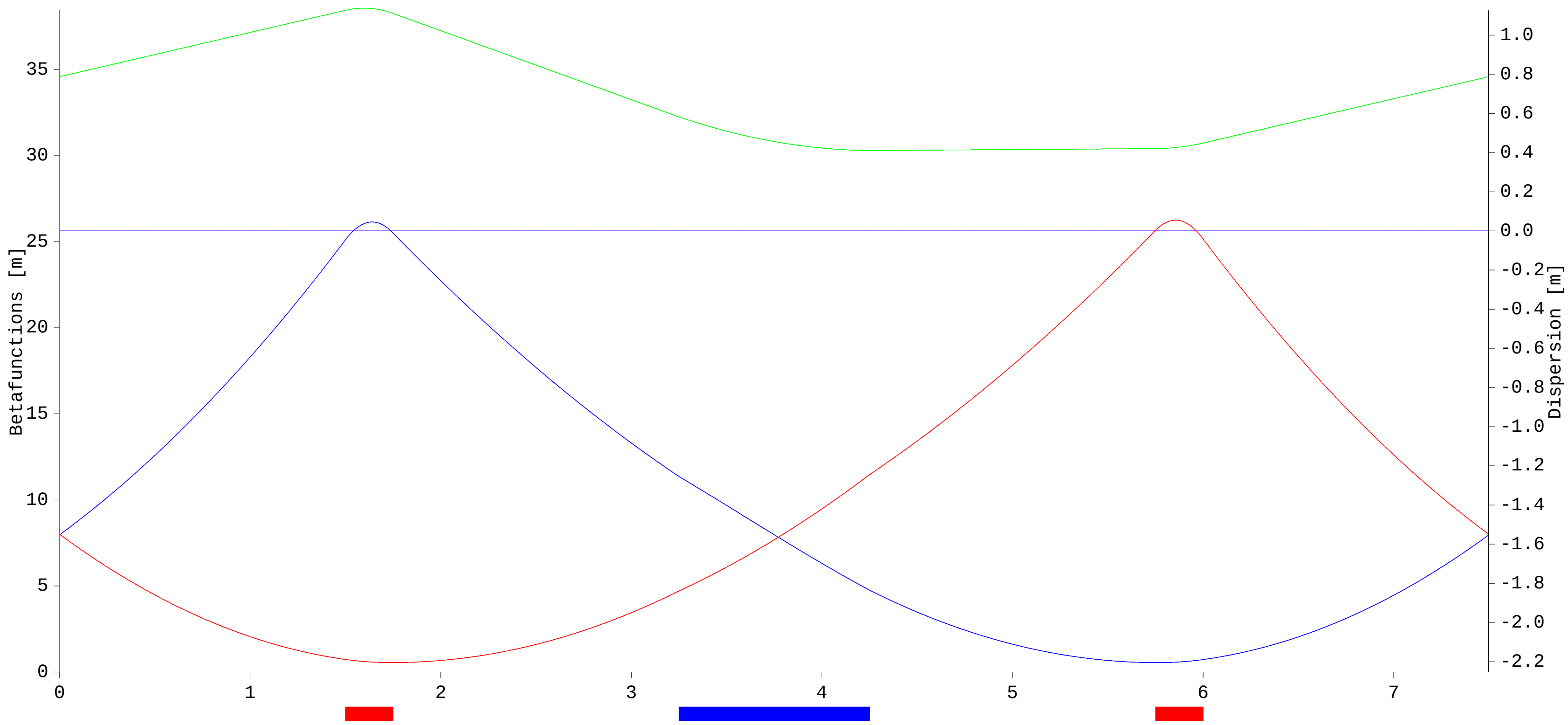
$$M_{V\ rectangular} = \begin{pmatrix} \cos\theta & \rho\sin\theta \\ -\frac{1}{\rho}\sin\theta & \cos\theta \end{pmatrix}$$

and for small angles, the matrices read:

$$M_{H\ rectangular} = \begin{pmatrix} 1 & \rho\theta & \frac{1}{2}\rho\theta^2 \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{pmatrix}$$

$$M_{V\ rectangular} = \begin{pmatrix} 1 & \rho\theta \\ -\frac{\theta}{\rho} & 1 \end{pmatrix}$$

FODO lattice

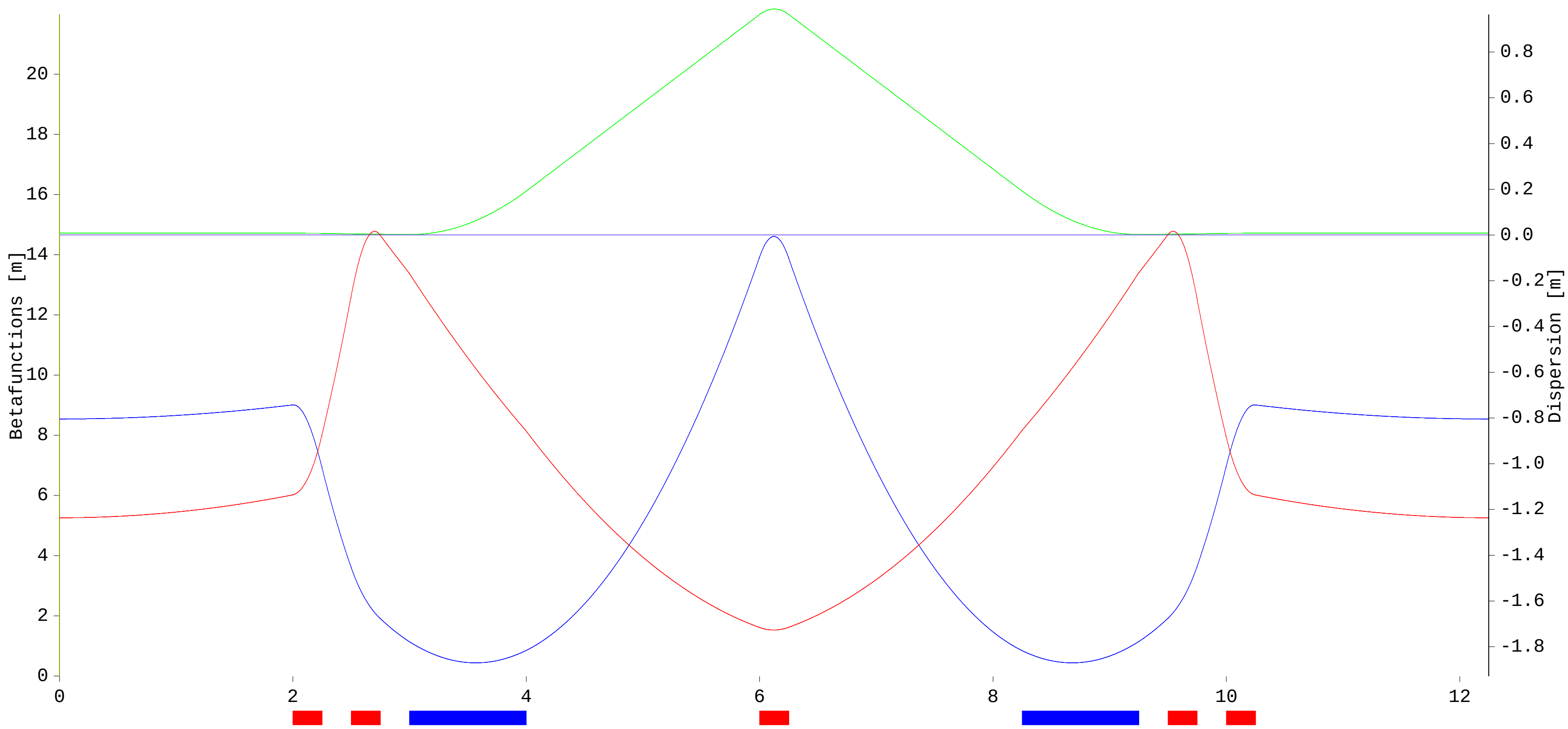


FODO Lattice

1) Crea una celda FODO y un anillo de 36 celdas para electrones de 2.5 GeV utilizando OPA (con el editor de texto o el editor visual incluidos en la aplicación).
   La secuencia es: drift + QF + drift + dipole + drift + QD + drift
   $L_{drift} = L_{dipole} = 1.2$ m, $L_{quad} = 0.25$ m, $k_{QF,QD} = \pm 1.2$ m^{-2}, y ángulos de entrada/salida del dipolo T1/T2 = 0.5* θ_{dipole} .
   Guarda el file de la lattice.
2) Cambia el valor de la k de QF hacia valores más grandes y más pequeños (pasos de ~0.2 m^{-2}), hasta que la solución se vuelve inestable (OPA no encuentra solución periódica). Observa y apunta cómo cambian los **tonos del anillo** en los dos planos y dibuja los dos tonos en función de k en una gráfica. Luego haz lo mismo con la k de QD.
   ¿Cuál cuadrupolo es más eficaz para cambiar el tono en cuál de los dos planos? Justifíca este resultado con las fórmulas físicas estudiadas en el curso.
3) Utilizando los resultados del ejercicio II de la lección E5, aplica en OPA un cambio de $\Delta k = +0.1$ m^{-2} en QF y el correspondiente Δk en QD para cambiar sólo Q_x y dejar invariado Q_y. Apunta los cambios de tonos realmente obtenidos con OPA, compáralos con los valores dados por la fórmula del ejercicio E5.II y valora si la fórmula aproxima bien o no los cambios de tono realmente producidos. Luego comprueba lo mismo con $\Delta k = -0.1$ m^{-2} en QD para cambiar solo Q_y y dejar invariado Q_x.
4) Ajusta las fuerzas de QF y QD para obtener la emitancia horizontal más baja posible, con la condición de que los máximos de las betas no pasen de 25 m. Con OPA haz la gráfica de las tallas del haz en el planos vertical y horizontal (“envelopes”) asumiendo un 2% de acoplamiento entre la emitancia vertical y horizontal. Apunta el valor de la emitancia horizontal, los valores de las k y de los tonos de tu lattice optimizada. Guarda las figuras de las funciones ópticas y de las tallas del haz para el informe.
5) Con los valores de las k fijados en el apartado anterior, cambia ahora el número de celdas del anillo a 18 y luego a 72 (ajustando el ángulo del dipolo correctamente). ¿Cómo cambian los valores máximos de las betas en los dos planos (aumentan, disminuyen, practicamente no cambian…), la dispersión máxima, la emitancia horizontal? Justifícalo con los conceptos estudiados en el curso y compáralo con el comportamiento previsto por las fórmulas que conoces.

DBA lattice

Double Bend Achromatic (DBA) cell:



DBA LATTICE (linear)

The purpose of this class is that each student builds his own storage ring lattice simulating with OPA two basic cells presented along the course: FODO, DBA. You will start working with the file named “dba_v1.opa”. For details about the use of the code, read the OPA User’s Guide (file named “opa.pdf”)

DBA cell: linear lattice (OPA: Optics Design panel)

1. Start with the theoretical QF2 *k* value (that assumes some approximations) to suppress the dispersion at the ends of the cell that you obtained in the DBA exercise (tutorial class E6). In OPA, adjust it by hand in small steps in order to obtain dispersion exactly zero. QF2 will keep this value all along the rest of this work.
2. Optimization: set the strengths of QF1 and QD1 to the same value as QF2 with the correct sign, play with the two k values and try to reduce as much as possible the horizontal emittance, again with a maximum beta in both planes lower than 25 m). Save the plots of the optimised betas and dispersion.
3. Write down a table with the minimum horizontal emittance you achieved and the relative maximum betas and dispersion, tunes (<u>for the whole ring</u>), chromaticity, *k* values. Compare the smallest emittance of your DBA ring with the smallest value for the FODO ring with the <u>same energy and number of dipoles</u>. Does the results of the comparison agree with what you expected from the concepts and formulas studied in the course? Explain why.
4. If you had to chose a position in the DBA cell where installing two chromatic sextupoles (i.e. sextupoles to correct the chromaticity), point out what could be the best positions for them to make the vertical and horizontal chromaticity zero and explain why.

DBA LATTICE (non-linear)

DBA: non-linear optics (OPA: Sextupoles and Phase Space panels)

The purpose of this exercise is performing basics simulations to correct the chromaticity of a DBA lattice with two families of sextupoles and simulating the dynamic aperture. Take the achromatic DBA ring you have optimized before and let's suppose that our vacuum chamber has round cross section with radius of 15 mm.

5. Calculate, with the theoretical formula introduced in the transverse dynamics classes, the natural <u>chromaticity of the whole ring</u> by picking up the quadrupole strengths k and the beta functions at the quad positions given by OPA.

6. Compare the natural chromaticity computed by OPA (that uses exact numerical solutions and not our analytical formulas) with the calculated one at point 5. Comment where the different result comes from.

7. Following the example in the input file, perform a particle tracking over 1024 turns starting at different horizontal amplitudes (e.g. start conditions: $x' = 0$ and x ranges from 0 to 15 mm in steps of 1 mm or more). Check that the movement is linear, i.e. the phase space trajectories are ellipses and the horizontal dynamic aperture (i.e. the transverse range in which particles are not lost) achieves the physical aperture of 15 mm. Save the plots of the phase space trajectories for the report.

8. Install two sextupoles SH and SV in the DBA cell to correct the chromaticity (see point 4). Notice: if you insert a sextupole in the OPA lattice defined with length L = 0, OPA takes it as a thin element with K equal to the integrated gradient along the magnet.

9. Calculate the sextupole strengths m to correct the chromaticity (*ChromX* and *ChromY*) to zero in both planes. For this, make use of the formula already presented in the course.

10. Set the sextupole strengths calculated at point 9 in your OPA lattice and check if the corrected chromaticity calculated by OPA is approximately zero. Notice: OPA uses a different definition for the sextupole strength m which includes a factor ½ with respect to the formula used in lessons L5, L12 and E6 (have a look at the OPA guide).

11. Perform again a tracking over 1024 turns at different horizontal amplitudes (e.g. in steps of 1 mm), check that at larger amplitudes the movement is not linear anymore and the phase space trajectories are distorted ellipses. Increasing the starting horizontal amplitude in steps of 0.5-1 mm until the particle gets lost, find the new horizontal dynamic aperture (i.e. the maximum amplitude with stable movement). Save the horizontal phase space tracking plot for the report.

12. If you should chose a position where inserting a harmonic sextupole, that does not change the chromaticity but only compensates the non-linearity introduced by SH and SV and improves the dynamic aperture: where would you try to install it and why?